

Eonyong Han

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Research Profile

I develop computational methods for transcriptome reconstruction, multi-omics integration, and immune-response interpretation. My dissertation connects computational performance with biological interpretation by examining how data selection and partitioning shape the evidence available for inference. My background combines computer science with experimental protein research.

Education

Ph.D., Computer Science and Engineering Sep. 2022 – Feb. 2027 (expected)

Kyungpook National University • Advisor: Prof. Inuk Jung

Dissertation: *Data Selection and Partitioning for Transcriptome Reconstruction, Multi-omics Integration, and Immune-response Interpretation*

M.S., Computer Science and Engineering 2020 – 2022

Kyungpook National University

Thesis: *A Study on Network Comparison with Invariant Topology Size Using Colon Cancer Data*

B.S., Biotechnology 2010 – 2019

School of Life Science and Biotechnology, Kyungpook National University

Research Experience

Graduate Researcher 2020 – Present

Computational Biology & Bioinformatics Lab, Kyungpook National University

Advisor: Prof. Inuk Jung

- Developed Deep-SemP, a semantic read-partitioning framework for parallel transcriptome assembly, using protein-derived representations and knowledge distillation to a nucleotide model.
- Conducted de novo transcriptome reconstruction and gene-expression analysis in *Cnidium officinale* under heat stress.
- Benchmarked ten clustering methods for multi-omics study design; developed pathway and network analyses for TCGA cancer data.
- Built random-forest and SHAP analyses of COVID-19 immune response; contributed to density-based analysis of SARS-CoV-2 mutation hotspots.

Research Intern, then Graduate Researcher Dec. 2018 – Aug. 2020

Structural Biology, School of Life Science and Biotechnology, KNU

Supervisor: Prof. Eric Di Luccio

- Produced recombinant proteins in *E. coli*; performed SDS-PAGE, western blotting, chromatographic purification, and vapour-diffusion crystallization.
- Contributed to an NRF-funded study of Set7 methyltransferase in *Schizosaccharomyces pombe* gametogenesis and its association with human infertility.

Publications

[†]Equal contribution. Author's name in **bold**.

Under review

1. **Han, E.[†]**, Park, J.[†], Kim, Y., & Jung, I. (2026). A deep semantic read-partitioning framework for parallel reference-free de novo transcriptome assembly in *Caenorhabditis*. Under review at *Bioinformatics*.
Code: github.com/cobi-git/Deep-SemP

Peer-reviewed journal articles

1. Shin, S.[†], **Han, E.[†]**, Seong, H., Kim, Y.I., Jung, I., & Jung, W. (2025). De novo transcriptome assembly and gene expression analysis of *Cnidium officinale* under high-temperature conditions. *BMC Genomics*, 26, 907.
[doi:10.1186/s12864-025-12051-5](https://doi.org/10.1186/s12864-025-12051-5)
2. **Han, E.[†]**, Kwon, H.[†], & Jung, I. (2025). A review on multi-omics integration for aiding study design of large scale TCGA cancer datasets. *BMC Genomics*, 26, 769.
[doi:10.1186/s12864-025-11925-y](https://doi.org/10.1186/s12864-025-11925-y)
3. Youn, S.[†], Jeong, D.[†], Kwon, H.[†], **Han, E.**, Kim, S., & Jung, I. (2025). Identification of severity related mutation hotspots in SARS-CoV-2 using a density-based clustering approach. *BioData Mining*, 18, 61.
[doi:10.1186/s13040-025-00476-3](https://doi.org/10.1186/s13040-025-00476-3)
4. **Han, E.**, Youn, S., Kwon, K.T., Kim, S.C., Jo, H.-Y., & Jung, I. (2024). Disease progression associated cytokines in COVID-19 patients with deteriorating and recovering health conditions. *Scientific Reports*, 14, 24712.
[doi:10.1038/s41598-024-75924-x](https://doi.org/10.1038/s41598-024-75924-x)
5. Jeon, J., **Han, E.Y.**, & Jung, I. (2023). MOPA: An integrative multi-omics pathway analysis method for measuring omics activity. *PLOS ONE*, 18(3), e0278272.
[doi:10.1371/journal.pone.0278272](https://doi.org/10.1371/journal.pone.0278272)

Technical Skills

- **Programming and tools:** Python, R, Bash, Git, LaTeX, Linux/HPC.
- **Genomics:** RNA-seq and differential expression, de novo transcriptome assembly, single-cell RNA-seq, variant/mutation analysis, miRNA–target analysis.
- **Pathways and networks:** KEGG/GO enrichment, WGCNA, pathway-unit correlation networks, graph edit distance, network propagation.
- **Machine learning:** Random forests, SHAP, density-based and consensus clustering, feature selection, benchmarking design.
- **Deep learning:** ESM-2, DNABERT-2, knowledge distillation, transformer fine-tuning, multi-GPU inference with PyTorch.
- **Assembly and computing:** Trinity, BUSCO, minimap2, StringTie, CD-HIT-EST, CAP3, Jellyfish; complexity-aware CPU scheduling and multi-server parallelization.
- **Data resources:** TCGA, GISAID, NCBI SRA, Ensembl, UniProt, KEGG; clinical EHR and cytokine-profile cohorts.

Funded Research Projects

Selected participation as a student researcher or research assistant. Principal investigators (PIs) are identified below.

Petascale in-storage computing for accelerating genomic DNA sequence analysis Sep. 2024 – Jun. 2026

NRF, Basic Research Laboratory • PI: Yongtae Kim

Quantum advantage for quantum-computing-based genomic DNA alignment Jun. 2025 – Dec. 2025

NRF • PI: Yongtae Kim

Time-series multi-omics integrative analysis and AI applications Apr. 2024 – Dec. 2025

KDCA • PI: Inuk Jung

Multi-omics integrative analysis and prognosis prediction modelling for COVID-19 patients Mar. 2022 – Dec. 2023

KDCA • PI: Inuk Jung

Development of an AI-based infection severity classification system Aug. 2022 – Dec. 2024

KHIDI • PI: Inuk Jung

Large-scale multi-omics data integration methods and open platform Mar. 2021 – Dec. 2021

NRF • PI: Inuk Jung

Machine learning methods for single-cell time-series transcriptome data Mar. 2021 – May 2022

NRF • PI: Inuk Jung

AI diagnostic technology for embedded imaging medical devices May 2022 – Dec. 2024

KEIT • PI: Jaeil Kim

High-performance spatiotemporal IoT data management and intelligence engine Mar. 2024 – May 2025

NRF • PI: Youngkyoon Seo

Usage-pattern classification for personalized air-conditioner operation May 2022 – Apr. 2023

LG Electronics, industry collaboration • PI: Inuk Jung

Role of Set7 in *Schizosaccharomyces pombe* gametogenesis and its association with human infertility Sep. 2019 – Aug. 2020

NRF • PI: Eric Di Luccio

Fellowship

BK21 FOUR Sep. 2020 – Aug. 2026

Intelligence-Convergence Software Education & Research Group

National Research Foundation of Korea • Participating graduate student

Reference

Prof. Inuk Jung Ph.D. Advisor

School of Computer Science and Engineering, Kyungpook National University

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